

Ashwini Mankari

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Data Engineer

PROFESSIONAL SUMMARY:

- Data Engineer with 5 years of experience designing, developing and optimizing scalable ETL/ELT pipelines, cloud platforms, enterprise data integration solutions across healthcare, retail, financial services domains.
- Hands-on expertise in AWS, Google Cloud Platform and Azure services including AWS Glue, S3, Lambda, Snowflake, BigQuery, Dataflow, Dataproc, Azure Data Factory, ADLS Gen2 and Synapse Analytics.
- Strong experience building batch and real-time data pipelines using Apache Spark, PySpark, Spark SQL, Apache Kafka, Pub/Sub, and event-driven architectures processing over 5 million events daily.
- Proficient in Python, SQL, PySpark and Shell Scripting for data transformation, workflow automation, orchestration, performance tuning and cloud-native data engineering implementations.
- Designed and implemented scalable data warehouse, lakehouse architectures using Snowflake, BigQuery, Synapse Analytics, dimensional modeling, Star Schema, Snowflake Schema methodologies supporting 250+ reporting tables.
- Developed enterprise data ingestion frameworks integrating structured and semi-structured data from APIs, databases, cloud storage, flat files, and streaming platforms processing 4+ TB of data daily.
- Experienced with Apache Airflow, Cloud Composer, Azure Data Factory, AWS Step Functions and Jenkins, automating 60+ production workflows and improving operational efficiency across production environments.
- Strong knowledge of data governance, lineage, metadata management, data quality frameworks, validation controls, and monitoring solutions ensuring trusted and compliant enterprise datasets.
- Optimized distributed Spark workloads, SQL queries and cloud-based data processing pipelines through partitioning, caching, indexing strategies, reducing processing times by 40% while improving resource utilization.
- Collaborate closely with business stakeholders, analysts and data scientists supporting advanced analytics, AI initiatives, machine learning workloads, CI/CD automation, Docker, Kubernetes and Agile delivery practices.

TECHNICAL SKILLS:

Programming Languages - Python, SQL, PySpark, Shell Scripting

Data Processing & Streaming - Apache Spark, PySpark, Spark SQL, Apache Kafka, Pub/Sub, Event-Driven Architecture

Cloud Platforms - AWS, Google Cloud Platform (GCP), Microsoft Azure

AWS Services - S3, Glue, Lambda, Redshift, Athena, EMR, IAM, CloudWatch, Step Functions, EC2

GCP Services - BigQuery, Cloud Storage, Dataflow, Dataproc, Pub/Sub, Cloud Composer, Cloud Functions, IAM

Azure Services - Data Factory, ADLS Gen2, Synapse Analytics, SQL Database, Key Vault, DevOps

Data Warehousing & Modeling - Snowflake, BigQuery, Azure Synapse Analytics, Redshift, Data Lake, Data Warehouse Architecture, Dimensional Modeling, Star Schema, Snowflake Schema

ETL / ELT & Orchestration - AWS Glue, Cloud Composer, Apache Airflow, Azure Data Factory, dbt, Jenkins

Databases & Storage - PostgreSQL, MySQL, SQL Server, Oracle, Cassandra, Neo4j

DevOps & CI/CD - Git, GitHub, Azure DevOps, Docker, Kubernetes, Terraform, CI/CD Pipelines

Data Quality & Governance - Data Validation, Data Lineage, Data Monitoring, Metadata Management, Data Governance

Reporting & Visualization - Power BI, Tableau

File Formats - Parquet, Delta Lake, JSON, CSV, Avro

AI & Advanced Analytics Exposure - AI-assisted Data Profiling, Anomaly Detection, Query Optimization

PROFESSIONAL EXPERIENCE:

Medtronic | Data Engineer

August 2024 – Present

Responsibilities:

- Designed and maintained 75+ AWS Glue and PySpark pipelines supporting ingestion, transformation, and delivery of healthcare datasets across clinical, operational, and compliance reporting domains.
- Processed over 4 TB of healthcare data daily from APIs, EHR systems, relational databases, and cloud storage platforms while maintaining SLA compliance and enterprise data quality standards.
- Led migration of enterprise healthcare analytics workloads from Amazon Redshift to Snowflake, modernizing the cloud data warehouse environment and consolidating over 250 reporting tables.

- Designed scalable Snowflake dimensional models and ELT frameworks using PySpark and SQL, improving query performance by 40% while reducing warehouse compute costs.
- Developed real-time ingestion pipelines using Apache Kafka, AWS Lambda, and Snowflake Snowpipe to process patient event, device telemetry, and operational datasets supporting near real-time analytics.
- Engineered Kafka-based streaming solutions processing over 5 million healthcare events daily, enabling faster visibility into operational metrics and critical healthcare reporting requirements.
- Automated ingestion and orchestration workflows using AWS Lambda and Step Functions, reducing manual intervention and improving reliability of recurring data processing workloads.
- Integrated structured and semi-structured datasets from APIs, databases, cloud storage, and third-party healthcare applications into centralized Snowflake analytics environments.
- Implemented Python and SQL-based data quality validation frameworks ensuring accuracy, consistency, governance compliance, and trusted delivery of healthcare datasets.
- Built metadata-driven ingestion frameworks accelerating onboarding of new data sources and increasing validation coverage by 35% across production pipelines.
- Monitored and optimized pipeline execution using CloudWatch, automated logging, and alerting mechanisms reducing incident response times and improving operational stability.
- Improved Spark processing efficiency through partitioning, caching, and performance tuning strategies reducing execution times from 4 hours to less than 90 minutes.
- Supported AI-driven healthcare initiatives by preparing governed datasets for machine learning, predictive analytics, and enterprise reporting solutions.

Lowe's | Data Engineer

August 2023 – August 2024

Responsibilities:

- Developed scalable ETL and ELT pipelines using Google Cloud Dataflow, Dataproc, and PySpark, processing retail sales, inventory, and customer datasets efficiently.
- Built real-time ingestion frameworks using Apache Kafka and Pub/Sub, enabling continuous processing of inventory updates, customer transactions, and operational events.
- Designed and optimized BigQuery warehouse solutions supporting enterprise reporting requirements across merchandising, customer analytics, supply chain operations, forecasting, and planning.
- Implemented Cloud Composer and Apache Airflow workflows orchestrating more than 60 production pipelines, improving operational reliability, scheduling efficiency and reducing manual effort.
- Developed PySpark transformation frameworks within Dataproc environments delivering analytics ready datasets supporting business intelligence, forecasting initiatives, performance analysis and strategic decisions.
- Automated data movement between Cloud Storage, BigQuery, operational databases, and downstream reporting systems, improving accessibility, integration efficiency, and enterprise availability.
- Designed dimensional models and curated data marts supporting executive reporting, inventory planning, customer analytics, self-service business intelligence, and decision-making.
- Implemented data quality validation, monitoring, lineage, and governance controls ensuring consistency, reliability, compliance, and trust across enterprise analytics datasets.
- Optimized BigQuery queries and Spark workloads reducing compute costs by 28%, while improving reporting performance, scalability, resource utilization, and processing efficiency.

Wipro (Client: HDFC Limited) | Data Engineer

September 2020 – December 2022

Responsibilities:

- Developed enterprise ETL pipelines integrating banking datasets from core financial systems into centralized analytics platforms, enabling reliable data consolidation, supporting regulatory reporting requirements.
- Designed scalable data ingestion frameworks for structured and semi-structured datasets, ensuring secure data movement, governance compliance and efficient enterprise-wide information accessibility.
- Built Spark-based transformation frameworks processing millions of customer, account and transaction records, supporting operational analytics, regulatory reporting, business-critical decision making processes.
- Implemented dimensional data models and reporting layers supporting lending, compliance, risk management, and operational reporting initiatives across multiple banking business functions.
- Automated recurring data integration workflows using Azure Data Factory, reducing operational effort by 22% while improving processing consistency, reliability and overall workflow efficiency.

- Integrated data from relational databases, APIs, flat files, and external banking applications into enterprise warehouse environments, improving accessibility, consistency, and reporting accuracy.
- Developed performance-tuned SQL queries, stored procedures, reconciliation processes and validation frameworks ensuring accurate, trusted and compliant banking datasets across enterprise platforms.
- Automated deployments through Azure DevOps CI/CD pipelines and collaborated with stakeholders to modernize reporting platforms, delivering scalable data engineering solutions aligned with business objectives.

EDUCATIONAL DETAILS:

Master of Science in Computer Science - University of Alabama, Birmingham

Bachelor of Technology in Electronics and Communication Engineering - Malla Reddy College of Engineering for Women